

LANDSCAPES OF THE CHURCH

LSAP Member Landscapes — 20 km & 100 km

Local ecological neighborhood, regional landscape context, member pairing, classified-cover correction, transition analytics, popup storytelling, QC, and code

LSAP Integral Ecology Atlas | Technical Documentation | 3 October 2026

12. LSAP Members — 20 km Local Ecological Neighborhood

12.1 Layer overview

Layer name	LandscapesChurch_Members_20km_LandCover_V2
Layer alias	LSAP Members - 20 km Local Ecological Neighborhood
Scale / features	Member point with 20 km analytical neighborhood; 2,739 features
Description	Member-centered land-cover layer describing the local ecological neighborhood around each publishable/geocodable LSAP member. It pairs local composition with the same member's 100 km regional landscape to create a local-to-regional story.
Attribution	Laudato Si' Action Platform publishable/geocodable member points; environmental summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons. Member points represent the consented/geocodable subset, not all LSAP registrations.
Citation / credits	Laudato Si' Action Platform / Dicastery for Promoting Integral Human Development, member data (project extract, Sep–Oct 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; project-accessed 2026 version).
Analyses	Eight-class composition; classified coverage; mosaic diversity/context; 20→100 km leading-landscape transition; largest compositional gains/losses; stewardship themes.
Key story fields	STORY_NEIGHBORHOOD; STORY_LOCAL_LANDSCAPE; STORY_LOCAL_TO_REGION; STORY_STEWARDSHIP; STORY_DATA_NOTE; STORY_SUMMARY; STORY_COMPREHENSIVE
Status	Publication-ready

12.2 Pairing and workflow

BUFFER_ID is the authoritative pairing key. Account_Name is not used as the join key because the same account name can appear more than once.

1. Preflight both 20 km and 100 km source tables and verify identical feature counts.
2. Audit candidate keys; select BUFFER_ID because it is non-null, unique (2,739/2,739), and identical across both scales.
3. Copy the source to derivative publication/analysis feature classes.
4. Calculate eight-class composition, coverage, leading pair, Shannon diversity, effective types, class counts, and mosaic context.
5. Diagnose and convert BUFFER_AREA and TOTAL_LULC_* from m² to km² once.
6. Separate Cloud+NoData from any additional unrepresented source coverage.
7. Recalculate mosaic context using the leading–second gap within normalized classified cover.
8. Pair local and regional records by BUFFER_ID and calculate normalized classified-composition changes.
9. Generate member-centered local-to-regional popup narratives and coverage-aware stewardship language.
10. Strip biodiversity-summary baggage, apply publication aliases, and add the layer to the top of the Landscapes of the Church group.

12.3 Key local-to-regional transition logic

Normalized class share at each scale	$s_i = 100 \times \text{raw_class_pct}_i / \text{VALID_COVER_PCT}$ Used only for cross-scale compositional change so missing classification does not masquerade as ecological change.
Regional change	$\Delta_i = s_{i,100\text{km}} - s_{i,20\text{km}}$ Positive values indicate a class becomes more prominent in the classified regional composition; negative values indicate it becomes less prominent.

BIGGEST_GAIN_TYPE / BIGGEST_GAIN_PP identify the largest positive Δ . BIGGEST_LOSS_TYPE / BIGGEST_LOSS_PP identify the most negative Δ . MAX_SHIFT_PP is the largest absolute change across the eight classes.

12.4 Transition reliability

- High-Confidence Comparison — minimum classified coverage across 20 km and 100 km $\geq 90\%$.
- Moderate-Confidence Comparison — minimum coverage $\geq 75\%$ and $< 90\%$.
- Limited-Coverage Comparison — minimum coverage $\geq 50\%$ and $< 75\%$.
- Very Limited-Coverage Comparison — minimum coverage $< 50\%$.

12.5 Core QC and final publication result

Retained QC output

MEMBER PAIRING KEY AUDIT

Account_Name

20 km: records=2,739 | nonnull=2,739 | unique=2,691 | duplicate values=47 | max duplicate=3
 100 km: records=2,739 | nonnull=2,739 | unique=2,691 | duplicate values=47 | max duplicate=3

BUFFER_ID

20 km: records=2,739 | nonnull=2,739 | unique=2,739 | duplicate values=0 | max duplicate=1
 100 km: records=2,739 | nonnull=2,739 | unique=2,739 | duplicate values=0 | max duplicate=1
 Values only in 20 km: 0
 Values only in 100 km: 0

PASS: exact 1:1 member pairing.

Retained QC output

20 KM AREA + COVERAGE AUDIT

Theoretical circular area: 1,256.637 km²
 Median BUFFER_AREA before conversion: 1,256,636,981.225
 Ratio if m²: 1.000000
 INTERPRETATION: BUFFER_AREA is square-meter scale.

Median summed TOTAL_LULC_*: 1,256,636,981.027
 Median TOTAL / BUFFER ratio: 1.000000

Classified coverage:

Minimum: 42.40%
 10th percentile: 95.58%
 25th percentile: 99.87%
 Median: 100.00%
 <90%: 140
 <80%: 21
 <70%: 2

All ten LULC percentages:

Median total: 100.000000%
 Minimum total: 99.999999%
 Maximum total: 100.000001%

Retained QC output

20 KM FINAL COVERAGE + MOSAIC SUMMARY

Coverage context:

Near-Complete Coverage	2,480
High Coverage	119
Moderate Coverage	136
Limited Coverage	3
Very Limited Coverage	1

Mosaic context:

Concentrated Landscape	363
Structured Mosaic	1,189
Diverse Mosaic	1,009
Highly Diverse Mosaic	178

Mosaic contexts changed after classified-gap correction: 9

Retained QC output

20 KM FINAL PUBLICATION CLEANUP

User fields before cleanup: 134
Fields identified for removal: 76
User fields after cleanup: 58
Required fields missing: 0
Story fields with missing values: 0
PASS

13. LSAP Members — 100 km Regional Landscape Context

13.1 Layer overview

Layer name	LandscapesChurch_Members_100km_LandCover_V2
Layer alias	LSAP Members - 100 km Regional Landscape Context
Scale / features	Member point with 100 km analytical neighborhood; 2,739 features
Description	Member-centered regional land-cover layer describing the broader landscape around each publishable/geocodable LSAP member and explicitly comparing it with the member's 20 km ecological neighborhood.
Attribution	Laudato Si' Action Platform publishable/geocodable member points; environmental summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons. Member points represent the consented/geocodable subset, not all LSAP registrations.
Citation / credits	Laudato Si' Action Platform / Dicastery for Promoting Integral Human Development, member data (project extract, Sep–Oct 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; project-accessed 2026 version).
Analyses	Eight-class composition; coverage; mosaic diversity/context; local→regional transition; regional stewardship context; coverage-aware story language.
Key story fields	STORY_REGION; STORY_REGIONAL_LANDSCAPE; STORY_LOCAL_TO_REGION; STORY_STEWARDSHIP; STORY_DATA_NOTE; STORY_SUMMARY; STORY_COMPREHENSIVE
Status	Publication-ready

13.2 100 km area interpretation

The median stored 100 km BUFFER_AREA was approximately 22,556.9 km² after conversion, smaller than the theoretical full-circle area of 31,415.9 km². The source TOTAL_LULC_* area sums closely matched the stored BUFFER_AREA, so the publication layer treats BUFFER_AREA as the analysis geography represented by the source workflow rather than claiming every feature is a mathematically complete 100 km circle.

13.3 Coverage model

The 100 km source required a more nuanced coverage model than the 20 km source. In most records, the eight valid classes plus Cloud and NoData sum to approximately 100%, but the lowest-coverage records revealed an additional residual of up to 15.40%. The workflow therefore distinguishes CLOUD_NODATA_PCT from OTHER_UNMAPPED_PCT while using UNCLASSIFIED_PCT = 100 – valid classified cover as the overall limitation.

13.4 Core QC

Retained QC output

100 KM AREA + COVERAGE AUDIT

Theoretical circular area: 31,415.927 km²
Median BUFFER_AREA before conversion: 22,556,872,817.134
Ratio if m² to full theoretical circle: 0.718008
INTERPRETATION: source analysis geography does not behave as a simple full theoretical circle.

Median summed TOTAL_LULC_*: 22,553,715,294.631
Median TOTAL / BUFFER ratio: 0.999860

Classified coverage:

Minimum: 8.74%
10th percentile: 64.97%
25th percentile: 82.35%
Median: 99.44%
<90%: 850
<80%: 639
<70%: 378

Cloud + NoData:

Median: 0.56%
Maximum: 83.82%

All ten LULC percentages:

Median total: 99.999994%
Minimum total: 84.603043%
Maximum total: 100.000000%
Maximum absolute error from 100%: 15.396957%

Retained QC output

100 KM FINAL COVERAGE + MOSAIC SUMMARY

Coverage context:

Near-Complete Coverage	1,633
High Coverage	256
Moderate Coverage	345
Limited Coverage	458
Very Limited Coverage	47

Mosaic context:

Concentrated Landscape	140
Structured Mosaic	874
Diverse Mosaic	1,090
Highly Diverse Mosaic	635

Maximum other unmapped coverage: 15.40%

Mosaic contexts changed after classified-gap correction: 80

Retained QC output

LOCAL -> REGIONAL TRANSITION DISTRIBUTION

Stable Leading Landscape	592
Same Leader, More Diverse Region	377
Same Leader, More Concentrated Region	169
Leading Landscape Changes	1,554
Coverage-Limited Transition	47

Comparison coverage:

High-Confidence Comparison	1,881
Moderate-Confidence Comparison	352
Limited-Coverage Comparison	459
Very Limited-Coverage Comparison	47

Most common leading shifts:

519	Built -> Trees
446	Trees remains leading
334	Built -> Cropland
315	Cropland remains leading
256	Built -> Rangeland
193	Rangeland remains leading
110	Water remains leading
105	Built remains leading
89	Built -> Water

Retained QC output

100 KM FINAL PUBLICATION CLEANUP

User fields before cleanup: 134
Fields identified for removal: 76
User fields after cleanup: 58
Required fields missing: 0
Story fields with missing values: 0
PASS

14. Cross-scale storytelling and popup design

14.1 Storytelling principles

- Put the place first. Popup prose should explain where the member or Church geography sits, not simply restate a table.
- Describe composition before interpretation.
- Avoid rank-heavy prose; keep rank fields analytical where useful but prefer direct landscape language.
- Do not describe bare land, snow/ice, concentrated composition, or high built share as inherently bad.
- Do not equate land-cover diversity with biodiversity, habitat quality, ecological health, or conservation value.
- Treat Water as surrounding physical geography and watershed/coastal context, not an assertion of ecclesiastical jurisdiction over water.
- When classified coverage is incomplete, explicitly say that the composition refers to the classified portion.
- At member scale, emphasize the transition from local 20 km neighborhood to wider 100 km region.

14.2 Member popup narrative model

The most distinctive storytelling innovation in the member layers is the local-to-regional transition. Instead of showing two independent percentage tables, the popup explains how the landscape changes as the spatial context expands.

Example pattern: “Built-up land leads around the member locally, while cropland becomes the leading land-cover type across the wider regional context. Within the classified landscape, cropland increases while built-up land decreases as the view expands.”

This makes the map actionable without prescribing a particular intervention. A locally urban institution can see its immediate heat/tree-canopy/stormwater context while also understanding its relationship to a broader agricultural, forest, rangeland, or watershed system.

14.3 Stewardship-language library

Leading type	Story themes
Built	urban heat; tree canopy; stormwater; habitat in developed areas; mobility; access to everyday nature
Trees	forest/woodland stewardship; habitat connectivity; canopy protection; watershed function; long-term carbon storage
Cropland	soil health; agroecology; pollinators; water quality; food systems; agricultural-community relationships
Rangeland	grassland/grazing systems; soil cover; drought resilience; connectivity; fire-aware stewardship
Water	watersheds; water quality; flood/coastal resilience; shoreline habitats; land-water relationships
Bare	erosion; water scarcity; fragile soils; arid-land ecology; stewardship suited to sparse vegetation
Snow/Ice	seasonal water systems; mountain/polar ecology; cryosphere dynamics; climate sensitivity
Other Vegetation	shrub/herbaceous habitats; fire ecology; soil condition; watershed function; connectivity

Relevant reproducible code

A.3 Shared land-cover helpers

```
import math

LANDCOVER = {
    "Water": "PCT_LULC_Water",
    "Trees": "PCT_LULC_Trees",
    "Cropland": "PCT_LULC_Crops",
    "Built": "PCT_LULC_Built",
    "Rangeland": "PCT_LULC_Rangeland",
    "Bare": "PCT_LULC_Bare",
    "Snow/Ice": "PCT_LULC_SnowIce",
    "Other Vegetation": "PCT_LULC_Veg"
}

def safe(v):
    return 0.0 if v is None else float(v)

def normalized_classified(raw):
    valid = sum(max(0.0, safe(v)) for v in raw.values())

    if valid <= 0:
        return valid, {k: 0.0 for k in raw}

    return (
        valid,
        {k: 100.0 * max(0.0, safe(v)) / valid for k, v in raw.items()}
    )

def shannon_metrics(raw):
    valid, pct = normalized_classified(raw)

    props = {k: v / 100.0 for k, v in pct.items()}
```

```

H = -sum(
    p * math.log(p)
    for p in props.values()
    if p > 0
)

diversity = 100.0 * H / math.log(8)
effective = math.exp(H)

meaningful = sum(v >= 1.0 for v in pct.values())
major = sum(v >= 5.0 for v in pct.values())

ordered = sorted(
    raw.items(),
    key=lambda kv: safe(kv[1]),
    reverse=True
)

top_type, top_pct = ordered[0]
second_type, second_pct = ordered[1]

classified_gap = (
    pct[top_type] -
    pct[second_type]
)

return {
    "valid_cover": valid,
    "diversity": diversity,
    "effective": effective,
    "meaningful": meaningful,
    "major": major,
    "top_type": top_type,
    "top_pct": safe(top_pct),
    "second_type": second_type,
    "second_pct": safe(second_pct),
    "classified_gap": classified_gap
}

def mosaic_context(m):
    if (
        m["diversity"] >= 65 and
        m["effective"] >= 4.0 and
        m["meaningful"] >= 5 and
        m["major"] >= 4 and
        m["classified_gap"] <= 15
    ):
        return "Highly Diverse Mosaic"

    if (
        m["diversity"] >= 55 and
        m["effective"] >= 3.25 and
        m["meaningful"] >= 4 and
        m["major"] >= 3 and
        m["classified_gap"] <= 35
    ):
        return "Diverse Mosaic"

    if (
        m["diversity"] < 35 or
        m["effective"] < 2.0 or
        m["major"] <= 1 or
        m["classified_gap"] >= 60
    ):
        return "Concentrated Landscape"

    return "Structured Mosaic"

```

A.10 Safe m² → km² conversion

```
import statistics
```

```
values = []
```

```

with arcpy.da.SearchCursor(
    feature_class,
    ["ANALYSIS_AREA"]
) as cursor:

    for (value,) in cursor:
        if value is not None and value > 0:
            values.append(float(value))

median_area = statistics.median(values)

if median_area >= 1_000_000:

    with arcpy.da.UpdateCursor(
        feature_class,
        ["ANALYSIS_AREA"]
    ) as cursor:

        for row in cursor:
            if row[0] is not None:
                row[0] = float(row[0]) / 1_000_000.0
                cursor.updateRow(row)

else:
    print("Area already appears to be in km²; no conversion performed.")

```

A.12 Member landscape foundation and coverage correction

```

# Eight valid landscape classes.
raw = {
    name: max(0.0, safe(row[field]))
    for name, field in LANDCOVER.items()
}

valid = sum(raw.values())

cloud = max(0.0, safe(row["PCT_LULC_Cloud"]))
nodata = max(0.0, safe(row["PCT_LULC_NoData"]))

cloud_nodata = cloud + nodata
all_ten = valid + cloud_nodata

other_unmapped = max(
    0.0,
    100.0 - all_ten
)

unclassified = max(
    0.0,
    100.0 - valid
)

# Diversity and type counts use only the classified portion.
m = shannon_metrics(raw)

MOSAIC_CONTEXT = mosaic_context(m)
VALID_COVER_PCT = valid
CLOUD_NODATA_PCT = cloud_nodata
OTHER_UNMAPPED_PCT = other_unmapped
UNCLASSIFIED_PCT = unclassified

```

A.13 Member 20 km ↔ 100 km pairing and transition

```

# BUFFER_ID is the authoritative pairing key.

def normalized_composition(raw_values, valid_cover):
    if valid_cover <= 0:
        return {
            name: 0.0
            for name in raw_values
        }

```

```

    return {
        name: value / valid_cover * 100.0
        for name, value in raw_values.items()
    }

# Example for one paired member:
local_norm = normalized_composition(
    local_raw,
    local_valid_cover
)

regional_norm = normalized_composition(
    regional_raw,
    regional_valid_cover
)

delta = {
    name:
        regional_norm[name] -
        local_norm[name]
    for name in LANDCOVER
}

gain_type = max(delta, key=delta.get)
loss_type = min(delta, key=delta.get)

gain_pp = delta[gain_type]
loss_pp = delta[loss_type]

max_shift_pp = max(
    abs(v)
    for v in delta.values()
)

diversity_change = (
    regional_diversity -
    local_diversity
)

leading_shift = (
    f"{local_top} remains leading"
    if local_top == regional_top
    else f"{local_top} -> {regional_top}"
)

```

A.14 Member comparison reliability and transition categories

```

def reliability(local_cov, regional_cov):
    minimum = min(local_cov, regional_cov)

    if minimum >= 90:
        return "High-Confidence Comparison"

    if minimum >= 75:
        return "Moderate-Confidence Comparison"

    if minimum >= 50:
        return "Limited-Coverage Comparison"

    return "Very Limited-Coverage Comparison"

def transition_context(
    local_top,
    regional_top,
    diversity_change,
    comparison_reliability
):
    if comparison_reliability == "Very Limited-Coverage Comparison":
        return "Coverage-Limited Transition"

    if local_top != regional_top:
        return "Leading Landscape Changes"

```

```

if diversity_change >= 8:
    return "Same Leader, More Diverse Region"

if diversity_change <= -8:
    return "Same Leader, More Concentrated Region"

return "Stable Leading Landscape"

```

A.15 Coverage-aware popup language pattern

```

def coverage_note(coverage, scale_word):
    if coverage >= 95:
        return ""

    if coverage >= 90:
        return (
            f"Land-cover classification covers {coverage:.1f}% "
            f"of the {scale_word} analysis geography."
        )

    if coverage >= 75:
        return (
            f"Land-cover information is available for {coverage:.1f}% "
            f"of the {scale_word} analysis geography, so the composition "
            f"described here refers to the classified portion."
        )

    if coverage >= 50:
        return (
            f"Land-cover information covers {coverage:.1f}% of the "
            f"{scale_word} analysis geography. The landscape interpretation "
            f"therefore describes only the classified portion."
        )

    return (
        f"Land-cover information is limited to {coverage:.1f}% of the "
        f"{scale_word} analysis geography. Landscape composition here "
        f"should be treated as a partial view."
    )

```

A.16 Publication cleanup pattern

```

# Keep only fields intentionally required for the publication derivative.
KEEP_FIELDS = {
    "BUFFER_ID",
    "Account_Name",
    "Country",
    "Language",
    "Official_website",
    "Sector",
    "BUFFER_AREA",
    "LANDSCAPE_SCALE",
    # ... eight PCT_LULC fields ...
    # ... coverage/mosaic/transition fields ...
    # ... story fields ...
}

delete_fields = [
    f.name
    for f in arcpy.ListFields(feature_class)
    if (
        not f.required and
        f.name not in KEEP_FIELDS
    )
]

if delete_fields:
    arcpy.management.DeleteField(
        feature_class,
        delete_fields
    )

# Always run a final required-field check after deletion.

```

Sources and attribution

LSAP member layers: Laudato Si' Action Platform / Dicastery for Promoting Integral Human Development. Public points represent the publishable/geocodable subset rather than all LSAP registrations.

Esri Global Biodiversity and Conservation Hexagons: <https://www.arcgis.com/home/item.html?id=f1bdbcac3c54ad8b5b21f5b17d91e8d>

Esri 2026 update: <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>

Summarize Within documentation: <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/analysis/summarize-within.htm>